

Graduate Seminar 2

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Section 1 - Solving for $\psi(x, t)$

- We solve $\psi(x, t)$ by a numerical resolution of the Time-Dependent Schrödinger Equation (TDSE), i.e

$$i \frac{\partial}{\partial t} \psi(x, t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \quad \psi(x, 0) = \psi_0(x), \quad (1)$$

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}}(t) = F(t)x. \quad (2)$$

$$E_0 = -\kappa^2/2, \quad \psi_0(x) = \kappa^{1/2} e^{-\kappa|x|}, \quad \kappa = 1, \quad (3)$$

$$E_k = k^2/2, \quad \psi_k^{(\pm)}(x) = e^{ikx} + R(\pm|k|)e^{\pm i|kx|}, \quad R(k) = \frac{i\kappa}{k - i\kappa}. \quad (4)$$

- The parameters of the laser we will be using are given by the relations

$$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t'), \quad 0 \leq t \leq T, \quad t' = t - T/2, \quad (5)$$

$$\tau = 2\pi n_{oc}/\omega_0, \quad n_{oc} = 5. \quad (6)$$

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Density, survival probability and PEMD

- The density probability of the wavefunction, at t is given by

$$\mathcal{D}(x, t) = |\psi(x, t)|^2 = \int \psi^*(x, t) \psi(x, t) dx. \quad (7)$$

- The coefficients of the projection of ψ are then, at t , and by definition, expressed as

$$a_0 = e^{iE_0 t} \int \psi_0(x) \psi(x, t) dx, \quad a_k = e^{iE_k t} \int \psi_k^{(-)*}(x) \psi(x, t) dx. \quad (8)$$

- Then,

$$P_0 = |a_0|^2, \quad \text{and} \quad P(k) = |a_k|^2. \quad (9)$$

- The ionization probability can then be found many different ways that should be consistent with each others

$$P_{\text{ion}} = \int P(k) \frac{dk}{2\pi} = 1 - P_0. \quad (10)$$

- $P(k)$ is the probability for the electron being ionized with the energy E_k .

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Math formula

- For an arbitrary operator $\mathcal{O}$, we have

$$\mathcal{O}(t) = \int \psi^*(x, t) \mathcal{O} \psi(x, t) dx \quad (11)$$

- The the Fourier transform is obtained using

$$\mathcal{O}(\omega) = \int \mathcal{O}(t) e^{i\omega t} dt. \quad (12)$$

- Following Bransden and Joachain, "Quantum Mechanics (2000)" (see 11.145), we define the transition amplitude $\mathcal{O}_{ba}$ and probability P_{ba} between two states a and b such as

$$\mathcal{O}_{ba} = \int dx \psi_b^*(x) \mathcal{O} \psi_a(x), \quad P_{ba} = \left| \int dx \psi_b^*(x) \mathcal{O} \psi_a(x) \right|^2. \quad (13)$$

When continuum states between q and $q + dq$ are involved, the quantity of interest is

$$P_{ba} dq = \left| \int dx \psi_b^*(x) \mathcal{O} \psi_a(x) \right|^2 dq. \quad (14)$$

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Length, velocity and acceleration

- We want to evaluate the quantities

$$d(t) = -\langle \hat{x} \rangle = -\int \psi^*(x, t) \hat{x} \psi(x, t) dx, \quad \text{and} \quad \dot{x}(t) = \frac{d}{dt} x(t), \quad (15)$$

$$p(t) = \langle \hat{p} \rangle = -i \int \psi^*(x, t) \frac{\partial}{\partial x} \psi(x, t) dx. \quad (16)$$

- Direct numerical differentiation can sometimes be avoided with the help of the Ehrenfest theorem and the commutator algebra

$$[\hat{p}^2, \hat{x}] = -2i\hat{p} \implies \hat{p} = i[\hat{H}, \hat{x}], \quad (17)$$

$$\frac{d}{dt} \langle \hat{x} \rangle = \langle \hat{p} \rangle, \quad \frac{d}{dt} \langle \hat{p} \rangle = \left\langle -\frac{\partial V_{\text{ext}}}{\partial x} \right\rangle. \quad (18)$$

and some others (see Bransden and Joachain, Quantum Mechanics (2000), p.531-532)

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High Harmonic Generation (HHG) - (semi-classical approx.)

- In the classical approx., the spectrum is obtained from the following (Eq.(80))

$$Q_{\nu}^{(c)}(\omega) = \alpha^2 \omega^2 |\mathbf{A}_{\nu} \mathbf{d}(\omega)|^2, \quad (19)$$

Let us apply the following simplifications

$$\nu \longrightarrow \omega, \quad \mathbf{A}_{\nu} = \sqrt{\frac{2\pi c^2}{\omega L^3}} \mathbf{e}_{\nu} \longrightarrow A_{\omega}, \quad \text{and} \quad \alpha A_{\omega} = 1. \quad (20)$$

- So the observable we are considering here is

$$Q^{(c)}(\omega) = \omega^2 |d(\omega)|^2. \quad (21)$$

Additionally let us consider Eq. (104)

$$Q^{(c)}(\omega) = |p(\omega)|^2. \quad (22)$$

and, by virtue of $\mathbf{p}(t) = -\dot{\mathbf{d}}(t)$, which reads

$$Q^{(c)}(\omega) = \left| -\dot{\mathbf{d}}(\omega) \right|^2. \quad (23)$$

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Additional formula from Ehrenfest theorem

- Using the integration by parts formula, we get

$$p(\omega) = \int_{-T/2}^{T/2} e^{i\omega t} \frac{dx(t)}{dt} dt, \quad u(t) = x(t), \quad v(t) = e^{i\omega t}, \quad (24)$$

thus

$$p(\omega) = \int_{-T/2}^{T/2} \frac{d}{dt} \left(e^{i\omega t} x(t) \right) dt - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt = \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt, \quad (25)$$

which reduces to (21) up to a constant.

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Transition amplitude

- From Yangaliev *et al.*, we have

$$p_{k0} = \int dx \psi_k^{(-)*}(x) \hat{p} \psi_0(x) = \frac{2k\mathcal{K}^{3/2}}{\mathcal{K}^2 + k^2}, \quad (26)$$

$$p_{kk'} = \int dx \psi_k^{(-)*}(x) \hat{p} \psi_{k'}^{(-)}(x) = 2\pi k \delta(k - k') - \frac{2\mathcal{K}}{k^2 - k'^2 + i\epsilon} \left(\frac{k'|k|}{|k| - i\mathcal{K}} - \frac{k|k'|}{|k'| + i\mathcal{K}} \right). \quad (27)$$

- Using (17), we can rewrite those expressions as

$$p_{k0} = -i(E_k - E_0) d_{k0}, \quad (28)$$

$$p_{kk'} = -i(E_k - E_{k'}) d_{kk'}. \quad (29)$$

- Those quantities are another way to study convergence, at the structural level

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Part 2 - Solving $\phi_\omega(x, t)$

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} S(x, t) \end{cases} \quad (30)$$

where

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}}(t) = F(t)x, \quad S(x, t) = \hat{p} \psi(x, t) \quad (31)$$

$$\phi_\omega(x, 0) = \frac{-i\kappa^{3/2}}{\omega} \text{sgn}(x) \left(e^{-\kappa|x|} - e^{-\kappa(\omega)|x|} \right), \quad \psi(x, 0) = \kappa^{1/2} e^{-\kappa|x|}, \quad (32)$$

$$\hat{p} = -id/dx, \quad \text{and} \quad \kappa(\omega) = \sqrt{\kappa^2 + 2\omega}. \quad (33)$$

$$\phi_\omega(x, t) = U(t, t_0) \phi_\omega(x, t_0) - i \int_{t_0}^t U(t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

High-order exponential splitting (composition methods) (Phys. Rev. A 101, 013410 (2020))

Yoshida (1990) - 4th-order symmetric decomposition

$$\phi_{\omega}(x, t + \Delta t) = U_4(t + \Delta t, t) \phi_{\omega}(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

- From the following Strang operator $U(h) = e^{-iE_n h/2} e^{-iV_{\text{ext}}(t+h/2)h} e^{-iE_n h/2}$, we compose

$$U_4(h) = U(c_1 h) U(c_2 h) U(c_1 h), \quad \text{with} \quad c_1 = \frac{1}{2 - 2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2 - 2^{1/3}}. \quad (35)$$

- More precisely, we can observe the following

Theorem 8.6 (Hairer, Solving ODE II p.554)

If Ψ_h represent a one-step method that is symmetric, of order $p = 2k$ and defined in the domain, then the relations

$$2c_1 + c_2 = 1, \quad 2c_1^{p+1} + c_2^{p+1} = 0, \quad (36)$$

imply that the composition (35) is of order $p + 2$.

If you start from the Strang method, then in the above $k = 1$.

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Midpoint Duhamel - 2nd order approximation

$$\phi_{\omega}(x, t + \Delta t) = U(t + \Delta t, t)\phi_{\omega}(x, t) - i \int_t^{t+\Delta t} U(t + \Delta t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

- We can show that the above can be expressed as

$$\phi_{n+1}^{\omega} = U(\Delta t)\phi_n^{\omega} - i\Delta t e^{i\omega t_{n+1/2}} S_{n+1/2}, \quad (37)$$

where the source is evaluated at $t_{n+1/2} = t + \Delta t/2$, i.e and $S_{n+1/2} = U(\Delta t/2)S_n$, with $S_n = S(x, t_n)$

- This is the midpoint Duhamel formula, of order 2.
- It's what will allow us to get familiar with the numerical intricacies of our problem
- The driver shall be regarded as decent if the expected order is met

Geometric propagators

Simpson Duhamel - 4th-order quadrature

$$\phi_\omega(x, t + \Delta t) = U_4(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

- The Simpson quadrature consists in writing

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left(U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \quad (38)$$

where the source is evaluated at t_n , $t_{n+\frac{1}{2}}$ and t_{n+1}

- This is the Simpson Duhamel formula, of order 4.
- It requires a 4th order evolution operator to work
- Simpson-Duhamel multiplies the number of operations required by Midpoint-Duhamel by 3.

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Observables

$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx, \quad (39)$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_\omega(x, T) dx, \quad (40)$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (41)$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (42)$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi} \quad (43)$$

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Midpoint Duhamel - Self conv. test results for $S(x, t) = \hat{p}\psi(x, t)$ - ZRP

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} \quad (38)$$

t	$\Re \psi(1)$	$\Im \psi(1)$	$\ \psi\ $	$\Re \phi(2)$	$\Im \phi(2)$	$\ \phi\ $
-1102.3132	1.0000000	0.0000000	1.0000000	0.0000000	1.4733146(-4)	0.6074294
-1080.2669	1.0000000	-1.5299150(-5)	1.0000000	9.0161878(-8)	1.4733146(-4)	2.9977570
-1058.2207	1.0000000	-3.0598299(-5)	1.0000000	1.8032375(-7)	1.4733145(-4)	4.2519438
			...			
-22.0463	-4.6878735(-7)	1.1266426(-6)	1.0000000	-3.4864304(-3)	-2.0577179(-3)	0.7712524
-0.0000	4.8643339(-1)	-9.9881227(-1)	1.0000000	-1.3514523	-8.0617898(-4)	3.3039060
22.0463	2.8495297(-7)	1.2366110(-6)	1.0000000	-3.7992762(-3)	1.3930728(-3)	4.2371529
			...			
1058.2207	-9.9526450(-1)	-9.7202282(-2)	1.0000000	1.5396368(-5)	-1.4659389(-4)	1.1029687
1080.2669	-9.9526599(-1)	-9.7187055(-2)	1.0000000	1.5389349(-5)	-1.4659255(-4)	3.5763755
1102.3132	-9.9526748(-1)	-9.7171829(-2)	1.0000000	1.5382713(-5)	-1.4659538(-4)	4.1762517

Level	Δt (relative)	E_ψ	E_ϕ	P_ψ/P_ϕ
1	Δt_0	0.1185025(-1)	0.1632911(+0)	—
2	$\Delta t_0/2$	0.2953778(-2)	0.3951342(-1)	2.004 / 2.048
3	$\Delta t_0/4$	0.7378959(-3)	0.9798974(-2)	2.001 / 2.011
Observed order				2.0011/2.0116

Ne = 100, Np = 10

-xmin = xrange/2 = -1333

$\tau = 2\pi n_{oc}/\omega_0$, $T = 8\tau$

Initial conditions and parameters

$S(x, t) = -i\partial_x \psi(x, t)$

$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t)$, $t' = t - T/2$

$F_0 = 0.03$, $\omega_0 = 0.114$, $\omega = 0.5$

$\Delta t_0 = 2.204$, $n_{oc} = 5$

■ Results of the time propagation for the zero range potential with and without source

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$

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Simpson Duhamel - Self conv. test results for $S(x, t) = \hat{p}\psi(x, t)$ - ZRP

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left(U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \quad (38)$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad (35)$$

t	$\Re \psi(1)$	$\Im \psi(1)$	$\ \psi\ $	$\Re \phi(2)$	$\Im \phi(2)$	$\ \phi\ $
-1102.3132	1.0000000	0.0000000	1.0000000	0.0000000	1.4733146(-4)	0.6074294
-1080.2669	1.0000000	-1.5299150(-5)	1.0000000	9.0161878(-8)	1.4733146(-4)	2.9977570
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-22.0463	-4.6878735(-7)	1.1266426(-6)	1.0000000	-3.4864304(-3)	-2.0577179(-3)	0.7712524
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Level	Δt (relative)	E_ψ	E_ϕ	p_ψ/p_ϕ
1	Δt_0	0.5564105(-3)	0.1529880(-1)	—
2	$\Delta t_0/2$	0.3480484(-4)	0.1093299(-2)	3.999 / 3.807
3	$\Delta t_0/4$	0.2175758(-5)	0.7047939(-4)	4.000 / 3.955
Observed order		3.9997/3.9553		

$$\text{Ne} = 100, \quad \text{Np} = 10$$

$$-x_{\min} = x_{\text{range}}/2 = -1333$$

$$\tau = 2\pi n_{oc}/\omega_0, \quad T = 8\tau$$

Initial conditions and parameters

$$S(x, t) = -i\partial_x \psi(x, t)$$

$$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t), \quad t' = t - T/2$$

$$F_0 = 0.03, \quad \omega_0 = 0.114, \quad \omega = 0.5$$

$$\Delta t_0 = 2.204, \quad n_{oc} = 5$$

■ Results of the time propagation for the zero range potential with and without source

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$

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Time dependence reference table - (homogeneous problem)

- Sufficient time step for converged result is $\Delta t \leq 0.08$ a.u.
- In the following, we will maintain Δt set to around 0.04 a.u.

Those are old data. Just a reminder

$\Delta t = 0.04$ a.u. $\omega = 0.057$ a.u. $\tau = 551.15661$									
$n = 3$		$n = 5$		$n = 8$		$n = 12$		$n = 16$	
$T = 3\tau$	nsteps	$T = 5\tau$	nsteps	$T = 8\tau$	nsteps	$T = 12\tau$	nsteps	$T = 16\tau$	nsteps
1653.4698	41336.745	2755.783	68894.576	4409.2528	110231.32	6613.8793	165346.98	8818.5057	220462.64

$\Delta t = 0.04$ a.u. $\omega = 0.114$ a.u. $\tau = 275.5783$									
$n = 3$		$n = 5$		$n = 8$		$n = 12$		$n = 16$	
$T = 3\tau$	nsteps	$T = 5\tau$	nsteps	$T = 8\tau$	nsteps	$T = 12\tau$	nsteps	$T = 16\tau$	nsteps
826.73491	20668.373	1377.8915	34447.288	2204.6264	55115.661	3306.9396	82673.491	4409.2528	110231.32

$$\tau = n_{oc} \frac{2\pi}{\omega}, \quad T = n\tau, \quad \Delta t = \frac{T}{\text{nsteps}}, \quad n_{oc} = 5,$$

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Convergence parameters table for the homogeneous equation

- The following table shows parameters known to return converged results for Eq. (1), which is recalled below

Those are old data. Just a reminder

What we want now, is to extend this logic to the inhomogeneous case

$\omega = 0.057 \text{ a.u.},$		$\Delta t \leq 0.08 \text{ a.u.},$	$\tau = 5 \times 2\pi/\omega = 551.15661 \text{ a.u.}$		
$F_0(\text{a.u.})$	3τ	5τ	8τ	12τ	16τ
0.03	(20/200) - 3000	(60/200) - 7000	(80/200) - 10000	(90/200) - 14000*	(90-300) - 16000*
0.1	(70/400) - 8000	(100/500) - 14000	?		
0.3	?				
1.0					

Legend

(nbas/nelm) - xmax

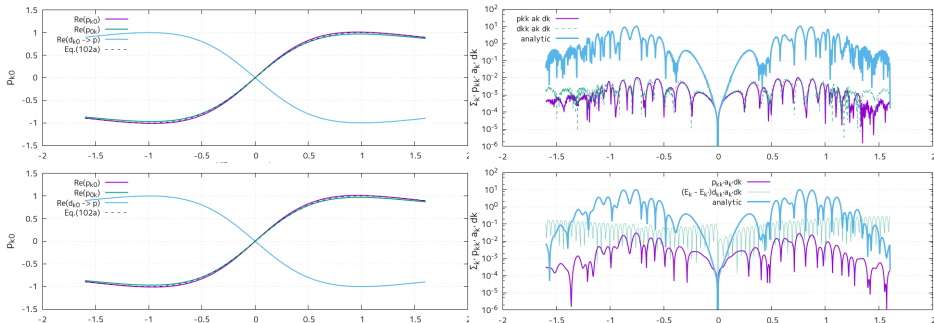
$\omega = 0.114 \text{ a.u.},$		$\Delta t \leq 0.08 \text{ a.u.},$	$\tau = 5 \times 2\pi/\omega = 275.5783 \text{ a.u.}$		
$F_0(\text{a.u.})$	3τ	5τ	8τ	12τ	16τ
0.03	(20/100) - 2000	(40/200) - 4000	(40/200) - 6000	(80/200) - 8000	(70-200) - 10000
0.1	(40/200) - 3000	(80/200) - 6000	(70/300) - 8000	(90/400) - 12000	(70-400) - 12000*
0.3	(90/400) - 7000	(100/500) - 10000			
1.0					

$$i \frac{\partial}{\partial t} \psi(x, t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \quad \psi(x, 0) = \psi_0(x),$$

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}} = F(t)x,$$

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Transition amplitudes



$N_p = 20, \quad N_s = 100$
 $x_{\max} \sim N_p N_s / 2$
 $\omega_0 = 0.114 \text{ a.u.}$
 $F_0 = 0.03 \text{ a.u.}, \quad T = 8\tau$

$$p_{k0} = \int dx \psi_k^{(-)*}(x) \hat{p} \psi_0(x) \quad (44)$$

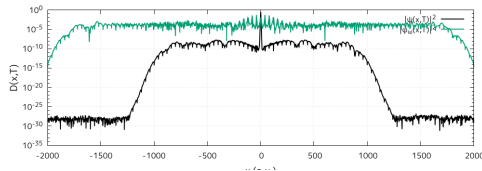
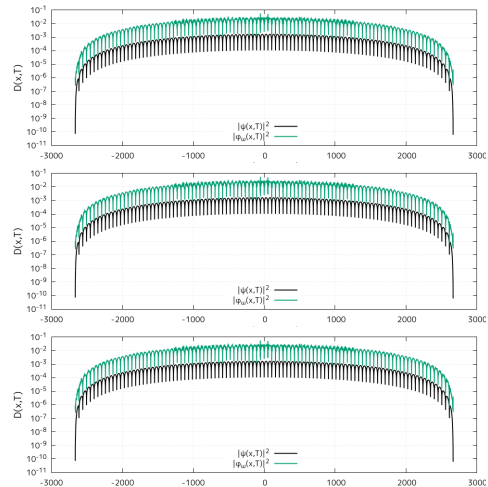
U: $N_p = 20, N_s = 100, x_{\max} = 1000$

D: $N_p = 40, N_s = 200, x_{\max} = 4000$

$$p_{kk'} a_k dk = \left(\int dx \psi_k^{(-)*}(x) \hat{p} \psi_{k'}^{(-)}(x) \right) a_k dk \quad (45)$$

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Density probability for the inhomogeneous case



Left, From up to down nsteps = 10k, 15k, 20k
Up $N_p = 20$, $N_s = 100$ $x_{\max} = 2000$

$$N_p = 20, \quad N_s = 100$$

$$x_{\max} \sim N_p N_s / 2$$

$$\omega_0 = 0.114 \text{ a.u.}$$

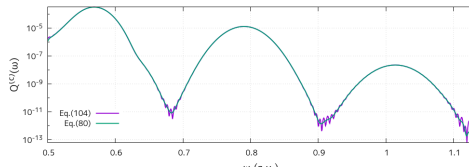
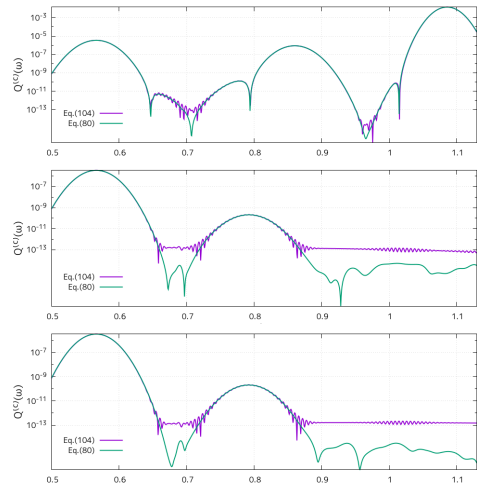
$$F_0 = 0.03 \text{ a.u.}, \quad T = 8\tau$$

The nonconvergence come from the fact that x here (46)

$$\mathcal{D}_\eta(x, t) = |\eta(x, T)|^2$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

HHG probability for the inhomogeneous case



Left, From up to down $nsteps = 10k, 1k, 100$
Up $N_p = 20, N_s = 200$ $x_{max} = 20$

$$N_p = 20, \quad N_s = 100$$

$$x_{max} \sim N_p N_s / 2$$

$$\omega_0 = 0.114 \text{ a.u.}$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 8\tau$$

The nonconvergence come from the fact that x_{max} is not large enough here

$$Q^{(c)}(\omega) = \omega^2 |d(\omega)|^2,$$

Above is Eq.80, below is Eq.(104)

$$Q^{(c)}(\omega) = |p(\omega)|^2.$$